Controllable Self Supporting Robotic Tail

Detailed development plan for a wearable multi section mechanism

75 hours • Python and Rust • ESP32 S3 • CAD and 3D printing • ROS 2

Project outcome

A safe bench-tested and wearable tail prototype approximately 70–90 cm long. It bends smoothly through space, forms arcs and S-curves, produces a travelling wave, and holds a commanded shape without manual tension. It is not intended to balance a person, support body weight, or prevent a fall.

Primary design decision

Use three serial active sections. Each section controls two bending directions through two closed tendon loops. The resulting six controlled degrees of freedom create diagonal, circular, wave-like, and expressive motion. An elastic backbone, tendon pretension, reduction, and feedback keep the mechanism from sagging uncontrollably.

Parameter	Target
Structure	3 active sections × 2 DOF
Vertebrae	9–12 printed elements
Control	gamepad, motion files, trajectory editor
Control loop	50–100 Hz on the microcontroller
Validation	bench → dummy/load → brief wearable test

Role in the programme. Complete this module after the single-joint and basic electronics work. It reuses sensor, control, embedded, CAD, and ROS 2 skills and prepares reusable components for the robotic hand and hexapod projects.

The 75 Hour Plan

Five blocks from requirements to demonstration

Block	Hours	Analysis	Practice	Result
1	15	requirements, risks, kinematics	models and calculations	Specification v1
2	15	CAD and tendon routing	one passive section	Section v0
3	15	actuation and embedded control	one active section	Section v1
4	15	trajectories and synchronisation	three-section bench	Tail bench v1
5	15	integration and validation	wearable prototype	Tail v1.0

Working rhythm

A two-hour session uses 10 minutes to restore context, 30 minutes for theory, 60 minutes for implementation, 15 minutes for measurements, and 5 minutes for the engineering log. A long session begins with a readiness criterion and ends with a repeatable test and saved evidence.

Repository layout

- requirements/ — scope, hazards, acceptance criteria
- cad/ — vertebra, section, mount, spool, STL and STEP
- firmware/ — drivers, watchdog and state machine
- python/ — shape simulator, motion editor and plots
- ros2_ws/ — messages, command nodes and telemetry
- tests/ — scenarios, expected results and reports
- motions/ — versioned YAML or JSON motion files

Progression rule. Do not start the next block until the current gate is closed. A visually attractive video does not replace measurements. Record the CAD revision, firmware commit, pretension, mass, command, plots, and observed defects.

Requirements and Project Boundaries

Define exactly what the prototype must and must not do

Functional requirements

- Bend smoothly in any direction within the validated working cone.
- Control the base, middle, and tip independently.
- Perform a diagonal arc, circle, S-curve, and travelling wave.
- Hold a static pose for at least 30 seconds within a stated tolerance.
- Move between poses without step changes in speed or tension.
- Stop on stale commands, excessive current, or communication loss.
- Never resume an old sequence automatically after a stop.

Area	Requirement	Verification
Mass	minimise moving mass	weigh each section and base
Noise	no gearbox impacts	record and review
Temperature	validated enclosure limit	sensor during a 10-minute cycle
Repeatability	pose repeats across trials	five identical runs
Repair	replace one tendon or vertebra	disassembly test

Exclusions. Version 1.0 does not balance the wearer, support body weight, act as an orthosis or medical device, operate while running, or perform high-energy swings. Add the decorative skin only after mechanical validation.

Kinematics of a Multi Section Tail

From two bending axes to a spatial curve

Represent one section with curvature components kx and ky, or with bending magnitude θ and bending-plane angle ψ. Combining the two components produces diagonal and circular motion. Connecting three sections produces a spatial curve without actuating every vertebra independently.

Command	kx	ky	Observed shape
Neutral	0	0	straight or pre-curved tail
Left	+	0	horizontal arc
Up	0	+	vertical arc
Diagonal	+	+	spatial arc
Circle	sin(ωt)	cos(ωt)	rotating bending direction

Travelling wave

For section i, define $q_i(t) = A_i \sin(\omega t + \varphi_i)$. A phase delay between the base, middle, and tip makes motion propagate along the tail. The tip may use a larger amplitude, but receives separate velocity and acceleration limits.

Required calculations

1. Define each local frame and positive bending direction.
2. Approximate the map from desired curvature to tendon-length change.
3. Validate neutral, principal axes, diagonal, and circle cases.
4. Animate the centreline in Python.
5. Compare predicted shape against a calibrated bench photograph.

Use a constant-curvature model for v1. It does not capture friction, backlash, or local deformation; physical calibration remains mandatory.

Vertebrae and Backbone Mechanics

A self-supporting structure without dangerous rigidity

Recommended structure

- 9–12 vertebrae with four tendon passages
- a flexible central rod or elastic backbone
- three sections of 3–4 vertebrae
- mechanical angle stops between neighbouring elements
- a soft tip and enclosed pinch zones
- a tool-free breakaway connection at the waist mount

Part	Prototype material	Reason
Vertebra	PETG	better repeated-load durability than PLA
Flexible insert	TPU 95A	return and damping
Backbone	fibreglass rod or spring strip	elastic load-bearing centreline
Tendon	Dyneema or similar braid	low stretch
Guide	PTFE liner	reduced friction on bends

CAD tasks

1. Create a parametric vertebra with diameter, pitch, tendon passage, and angle stop.
2. Remove sharp edges and prevent passage intersections.
3. Print three pitch variants and measure minimum bend radius.
4. Design a replaceable vertebra connection.
5. Publish an assembly drawing and component mass table.

Tendon Transmission

Spatial bending and shape retention

Each active section uses two perpendicular closed tendon loops. Rotation of a spool shortens one branch while lengthening the opposite branch. Moving both loops together aims the bending plane between the principal axes.

Pretension

A tendon must not leave its guide when direction reverses. Modest pretension removes slack; a spring or tensioning roller absorbs tolerances. Excessive pretension increases friction, heating, and base load.

Defect	Symptom	Correction
Loose tendon	delay and snap after reversal	set pretension and recalibrate zero
Excess pretension	high current and poor return	reduce tension
Friction	different shape by approach direction	increase radius and improve routing
Stretch	pose slowly drifts	change tendon or add feedback
Cross-coupling	one section moves another	secure and separate guides

Mechanical gate

- One passive section bends continuously without binding.
- No wear or cracks after 100 manual cycles.
- Neutral repeats after unloading.
- Stops prevent backbone over-bending.
- The section carries its own mass in a test pose.

Actuators and Pose Holding

Prevent sag without continuous overheating

Self-support comes from the combined backbone stiffness, reduction, tendon pretension, and position feedback. The backbone carries part of the weight; tendons define curvature; the actuator supplies the remaining moment.

Option	Advantages	Limitations	Use
Servo	fast start, integrated position	noise, backlash, limited diagnostics	early bench
DC gearmotor	current and velocity control	needs encoder and driver	main prototype
Stepper	simple holding	heat and missed steps	bench spool
Worm gear	mechanical holding	friction and poor backdrivability	only after risk review

Calculate before buying

Estimate distal mass, lever arm, required tendon force, and spool radius for each section. Approximate spool torque as $T = F \cdot r$. Add a documented allowance for friction and dynamics; do not select a motor from advertising peak torque alone.

Holding without overheating

- bias the neutral shape with an elastic element
- reduce moving mass toward the tip
- use reduction to lower holding current
- limit time in demanding static poses
- measure current and temperature

Electronics and Power

Six channels with an independent stop path

Node	Purpose
ESP32 S3	real-time control, commands, telemetry, watchdog
Motor drivers	direction, PWM, current or fault feedback
Encoders	spool or shaft position for every channel
Limit switches	homing and hard travel boundary
Fused supply	limited and measurable bench energy
E stop	independent removal of motor power

Wiring and measurements

- Separate power and signal wiring and document grounding.
- Place decoupling close to motor drivers.
- Log supply voltage, current, position, and stop reason.
- Measure voltage sag when several channels start together.
- Keep batteries and exposed conductors away from the body during early tests.

Commissioning order

1. One motor without a tendon and with a low current limit.
2. One motor with spool and mechanical stop.
3. One section with two actuators.
4. Three sections on a fixed bench.
5. Only then portable power and a wearable mount.

Embedded Software

States, control loops, and safe stopping

State	Permitted action	Exit
BOOT	self-test, outputs disabled	READY or FAULT
READY	accept configuration	HOMING or STOPPED
HOMING	slow zero search	STOPPED or FAULT
MANUAL	accept new position commands	STOPPED or FAULT
PLAYBACK	execute one trajectory	STOPPED or FAULT
STOPPED	clear motion and hold safely	explicit reset only
FAULT	disable power output	manual diagnosis and reset

Control layers

- The outer loop commands section curvature or pose.
- The inner loop controls spool position and limits velocity.
- The limiter checks current, temperature, angle, and command rate.
- The watchdog enters STOPPED on a stale command.
- A duplicate packet does not refresh command freshness.

Rust tasks

1. Model states and legal transitions with enums.
2. Use monotonic timestamps and command-age checks.
3. Store telemetry in a fixed-capacity ring buffer.
4. Test STOPPED, FAULT, and restart transitions.
5. Preserve the first fault cause.

Smooth and Expressive Motion

Trajectory quality matters more than motor count

Flexible mechanics will still move abruptly when commanded with steps. The planner limits position, velocity, acceleration, and preferably jerk. Host software plans in section coordinates; embedded software executes spool coordinates.

Motion	Sections	Principle
Sway	1–3	small sine waves with phase delay
Circle	1–2	sine and cosine on two axes
S-curve	1 and 2 opposite	static spatial curvature
Wave	1 → 2 → 3	one frequency and delayed phase
Curious tip	3	small arc over the base pose
Return	all	velocity-limited transition to neutral

Python editor

- timeline with key poses
- cubic interpolation or S-curve profile
- 3D centreline preview
- position, velocity, and acceleration plots
- versioned export with frequency and units

Machine learning is not required for v1. Motion must first be deterministic, explainable, and constrained. Learning from demonstration belongs after reliable telemetry and a validated limiter.

ROS 2 Architecture

Separate planning from real-time execution

Node	Input	Output
tail_input	gamepad	desired pose and mode
tail_planner	pose or motion ID	timed trajectory
tail_limiter	trajectory	safe command
tail_bridge	command	serial or CAN frames
tail_state	telemetry	joint state and faults
tail_visualizer	state	RViz markers and plots

Command contract

Every command contains a sequence ID, monotonic timestamp, mode, six target coordinates, limits, and expiry. Units and signs are part of the interface. Reject unknown schema versions and incorrect channel counts.

Mode switching

- MANUAL cancels playback through an explicit user action.
- A new trajectory starts from measured state.
- STOPPED clears queued motion.
- The embedded watchdog stops locally after ROS 2 loss.
- RViz displays command and measurement separately.

Calibration and Bench Validation

Measure first and wear later

Calibration sequence

1. Find mechanical neutral with unloaded tendons.
2. Set equal pretension with a documented method.
3. Home each spool at limited velocity.
4. Map spool travel to section curvature on a test grid.
5. Measure hysteresis from both approach directions.
6. Version coefficients with the physical assembly.

ID	Scenario	Saved data	Pass criterion
T01	neutral for 60 s	position, current, temperature	drift within tolerance
T02	six boundary poses	command and measurement	no contact with hard stop
T03	100 cycles	current, events, tendon photos	no damage
T04	travelling wave	six channels and video	no steps or slack
T05	communication loss	event log	transition to STOPPED
T06	soft jam fixture	current and fault	rapid shutdown

Simulate a jam with a low-energy bench fixture. Never restrain a powered mechanism by hand and never introduce a deliberate fault while it is worn.

Wearable Integration and Safety

The tail moves itself but must not move the wearer

Mount

- A wide belt or rigid frame distributes load.
- The motor block stays close to the body and inside a guard.
- The tail detaches without tools.
- Wiring cannot form loops near the legs.
- Both wearer and observer can reach a stop control.

Hazard	Risk reduction	Verification
Tail impact	soft tip, velocity limit, clear zone	bench test
Pinch point	closed skin and controlled gaps	inspection in every pose
Overheating	sensor and current/time limits	10-minute cycle
Tendon failure	guard and safe neutral	controlled failure test
Unexpected start	manual reset and watchdog	power-cycle test
Lower-back load	low mass and short trials	mount assessment

Wearable test order. Begin stationary at minimum velocity with one channel, then one section, then one slow full-tail sequence. Stop immediately for pain, numbness, rubbing, unstable mounting, or unusual heating.

Purchasing and Tools

Buy only what the current gate requires

Stage	Required	Note
Available	Creality CR-10 SE	print vertebrae and spools
Block 1	caliper, scale, fasteners, heat-set inserts	measurement and assembly
Block 2	PETG, TPU 95A, tendon, PTFE guides	passive section
Block 3	ESP32-S3, two motors, drivers, encoders	one section first
Block 3	fused bench supply and E stop	bench safety
Block 4	four more motors and driver channels	after the one-section gate
Block 5	waist frame, guards, soft tip	after bench validation

Install

- Ubuntu 24.04, Git, and an editor
- Python, uv, NumPy, SciPy, Matplotlib, and pytest
- Rustup, Cargo, and ESP32 target tools
- ROS 2 and RViz for integration
- FreeCAD or Fusion 360
- KiCad after breadboard validation
- logic analyser and multimeter

Do not buy six expensive actuators at once. Measure the force and motion quality of one active section, then select a consistent motor family from actual data.

Acceptance Terms and Reading

What must be demonstrated at the end

Final gate

- All three sections independently pass the calibration grid.
- The tail performs neutral, diagonal, circle, S-curve, and wave motions.
- A static pose is held for 30 seconds within tolerance.
- No tendon, vertebra, or mount damage appears after 100 cycles.
- Communication loss, overcurrent, and E stop produce a verified stop.
- No motion begins after power cycle without explicit authorisation.
- CAD, BOM, firmware, Python tools, ROS 2 interfaces, logs, and video are retained.

Term	Definition
Degree of freedom	independent configuration coordinate
Curvature	rate of centreline direction change
Antagonistic drive	opposing force pair
Pretension	initial tension that removes slack
Hysteresis	path-dependent response to the same command
Watchdog	freshness monitor that enters a safe state

Short reading list

- Lynch and Park, Modern Robotics.
- Spong, Hutchinson, and Vidyasagar, Robot Modeling and Control.
- Webster and Jones, Design and Kinematic Modeling of Constant Curvature Continuum Robots.
- Camarillo et al., Mechanics Modeling of Tendon Driven Continuum Manipulators.
- Arque, Artificial Biomimicry Inspired Tail for Extending Innate Body Functions, ACM SIGGRAPH 2019.
- ROS 2, ESP-IDF, and The Embedded Rust Book documentation as required by each block.

Session 1 Measurable Requirements

Hours 1–5 • Block 1

Session outcome

A one-page specification v0.1 with measurable criteria, explicit exclusions, and five reference motions.

Theory 60 minutes

- scope and v1 boundaries
- configuration, state, and degree of freedom
- requirement, metric, tolerance, and test method
- verifiable behaviour versus visual impression

Practice 180 minutes

1. Describe the stationary-use scenario.
2. Set tentative length, section count, and working cone.
3. Draw the three sections and local axes.
4. Specify five motions from start pose to completion.
5. Turn smoothness and pose holding into measurable tests.

Measurement and report 60 minutes

Retain	Purpose
requirements_v01.md	evidence of completion and input to the next session
axes sketch	evidence of completion and input to the next session
acceptance table	evidence of completion and input to the next session
exclusions list	evidence of completion and input to the next session

Completion criterion. Every requirement has a test method; balancing the wearer is explicitly excluded.

Typical mistakes. Do not freeze exact motor specifications before the first mechanical experiment.

Session 2 Risk and Safe Architecture

Hours 6–10 • Block 1

Session outcome

A hazard register and stop architecture created before powered mechanics.

Theory 60 minutes

- hazard, severity, probability, and residual risk
- safe state
- unexpected-start prevention
- energy limitation

Practice 180 minutes

1. List impact, pinch, tendon failure, heating, and link-loss hazards.
2. Define prevention, detection, and response for each hazard.
3. Choose STOPPED behaviour.
4. Draw an independent E-stop power path.
5. Define the order of bench and wearable tests.

Measurement and report 60 minutes

Retain	Purpose
risk_register.csv	evidence of completion and input to the next session
safe_state.md	evidence of completion and input to the next session
E-stop diagram	evidence of completion and input to the next session
test_order.md	evidence of completion and input to the next session

Completion criterion. No single software fault can start motion after a stop or power-up.

Typical mistakes. Do not rely on ROS 2 as the only protection layer.

Session 3 Kinematic Model in Python

Hours 11–15 • Block 1

Session outcome

A three-section model, 3D preview, and tests for known shapes.

Theory 60 minutes

- constant curvature
- kx and ky
- local and world frames
- transform composition

Practice 180 minutes

1. Implement one section from length and curvature.
2. Chain three sections.
3. Plot neutral, four directions, and diagonal.
4. Animate a rotating bending plane.
5. Test zero curvature, continuity, and symmetry.

Measurement and report 60 minutes

Retain	Purpose
tail_model.py	evidence of completion and input to the next session
test_kinematics.py	evidence of completion and input to the next session
five plots	evidence of completion and input to the next session
units_and_frames.md	evidence of completion and input to the next session

Completion criterion. All reference shapes are reproducible and units are documented.

Typical mistakes. Do not mix degrees and radians or compare plots without saved parameters.

Session 4 Parametric Vertebra

Hours 16–20 • Block 2

Session outcome

Three CAD variants and one printed test chain.

Theory 60 minutes

- parametric dependencies
- FDM tolerances
- guide radius and friction
- stress concentration

Practice 180 minutes

1. Create diameter, pitch, four passages, and centre-hole parameters.
2. Round tendon entry and exit edges.
3. Add a relative-angle stop.
4. Export three pitch variants.
5. Print a short chain and test tendon travel.

Measurement and report 60 minutes

Retain	Purpose
CAD sources	evidence of completion and input to the next session
STL and STEP	evidence of completion and input to the next session
print profile	evidence of completion and input to the next session
variant photos	evidence of completion and input to the next session
dimension table	evidence of completion and input to the next session

Completion criterion. The tendon moves freely and the vertebrae meet the designed stop first.

Typical mistakes. Do not size the passage at the tendon nominal diameter or leave sharp printed edges.

Session 5 Passive Section

Hours 21–25 • Block 2

Session outcome

A passive section with backbone, return behaviour, and manual tendon control.

Theory 60 minutes

- elastic-beam bending
- neutral shape
- stiffness and damping
- holding versus compliance

Practice 180 minutes

1. Assemble 3–4 vertebrae.
2. Route four temporary tendons.
3. Measure mass, length, and minimum bend radius.
4. Run ten manual cycles in four directions.
5. Compare three backbone variants.

Measurement and report 60 minutes

Retain	Purpose
section_v0.step	evidence of completion and input to the next session
mass and dimensions	evidence of completion and input to the next session
cycle video	evidence of completion and input to the next session
backbone comparison	evidence of completion and input to the next session

Completion criterion. The section returns near neutral, does not bind, and carries its own mass.

Typical mistakes. Do not make the section dangerously rigid merely to prevent sag.

Session 6 Spool and Closed Tendon Loop

Hours 26–30 • Block 2

Session outcome

A manual two-axis section rig with repeatable pretension adjustment.

Theory 60 minutes

- antagonistic pair
- tendon-length change
- spool radius and resolution
- Bowden hysteresis

Practice 180 minutes

1. Design a spool that anchors both loop ends.
2. Add tension adjustment without retying knots.
3. Build two perpendicular loops.
4. Mark zero and angular scale.
5. Measure shape from both approach directions.

Measurement and report 60 minutes

Retain	Purpose
spool_v1.step	evidence of completion and input to the next session
routing diagram	evidence of completion and input to the next session
pretension record	evidence of completion and input to the next session
hysteresis plot	evidence of completion and input to the next session

Completion criterion. Both axes combine into a diagonal pose without creating slack.

Typical mistakes. Do not allow uncontrolled crossing layers on the spool.

Session 7 One Actuator Channel

Hours 31–35 • Block 3

Session outcome

One safe motor channel tested before attachment to the full tail.

Theory 60 minutes

- torque, speed, and power
- encoder and gear ratio
- PWM and direction
- current as a diagnostic

Practice 180 minutes

1. Power one driver with a low current limit.
2. Verify direction without a tendon.
3. Read the encoder and define sign.
4. Add software and mechanical limits.
5. Measure current at idle, acceleration, and stop.

Measurement and report 60 minutes

Retain	Purpose
channel schematic	evidence of completion and input to the next session
motor_test firmware	evidence of completion and input to the next session
position-current log	evidence of completion and input to the next session
motor data sheet	evidence of completion and input to the next session

Completion criterion. The channel tracks a small position command and stops safely on link loss.

Typical mistakes. Never apply full power before validating feedback sign.

Session 8 One Active Two Axis Section

Hours 36–40 • Block 3

Session outcome

Section v1 with two actuators, four directions, and diagonal motion.

Theory 60 minutes

- independent position loops
- mechanical cross-coupling
- saturation and anti-windup
- command-rate limiting

Practice 180 minutes

1. Connect one actuator to one closed loop.
2. Tune low-speed position control.
3. Add the perpendicular loop.
4. Run a 3 × 3 command grid.
5. Measure current at neutral and boundary poses.

Measurement and report 60 minutes

Retain	Purpose
section_v1 assembly	evidence of completion and input to the next session
controller settings	evidence of completion and input to the next session
nine test poses	evidence of completion and input to the next session
command-position plots	evidence of completion and input to the next session

Completion criterion. The section completes the pose grid without snap, bind, overcurrent, or slack.

Typical mistakes. Tune one loop at a time; do not hide mechanical defects with aggressive PID.

Session 9 Firmware and Fault Transitions

Hours 41–45 • Block 3

Session outcome

A tested state controller with watchdog, fault latch, and homing.

Theory 60 minutes

- finite-state machine
- monotonic time
- latched fault
- command-state-event separation

Practice 180 minutes

1. Implement BOOT through FAULT states.
2. Disable power until successful initialisation.
3. Add command-freshness timeout.
4. Simulate duplicate, loss, and range faults.
5. Verify that power cycling does not start motion.

Measurement and report 60 minutes

Retain	Purpose
state_machine.rs	evidence of completion and input to the next session
transition table	evidence of completion and input to the next session
unit tests	evidence of completion and input to the next session
events.json	evidence of completion and input to the next session
E-stop video	evidence of completion and input to the next session

Completion criterion. Every failure reaches a known state and preserves the first cause.

Typical mistakes. Never auto-clear FAULT or refresh the watchdog with a duplicate command.

Session 10 Three Section Mechanical Assembly

Hours 46–50 • Block 4

Session outcome

A full tail on a fixed bench with routed tendons and measured mass.

Theory 60 minutes

- cascaded section load
- distal mass and base moment
- guide routing
- module repairability

Practice 180 minutes

1. Connect three sections.
2. Route six loops without crossings or sharp bends.
3. Mount actuators on a rigid bench.
4. Weigh each section and the complete tail.
5. Check manual travel before power.

Measurement and report 60 minutes

Retain	Purpose
tail_bench_v1.step	evidence of completion and input to the next session
mass BOM	evidence of completion and input to the next session
routing plan	evidence of completion and input to the next session
boundary-pose photos	evidence of completion and input to the next session

Completion criterion. Each actuator affects only its intended coordinate within the documented coupling limit.

Typical mistakes. Do not wear this assembly or increase tension to mask bad routing.

Session 11 Six Channel Calibration

Hours 51–55 • Block 4

Session outcome

A versioned map from section commands to spool positions.

Theory 60 minutes

- zero and homing
- coupling matrix
- repeatability and reproducibility
- data-based backlash compensation

Practice 180 minutes

1. Calibrate each neutral independently.
2. Collect a command grid per section.
3. Repeat points from positive and negative directions.
4. Estimate cross-coupling.
5. Bind coefficients to the mechanical assembly revision.

Measurement and report 60 minutes

Retain	Purpose
calibration_v1.yaml	evidence of completion and input to the next session
raw CSV	evidence of completion and input to the next session
hysteresis plots	evidence of completion and input to the next session
calibration report	evidence of completion and input to the next session

Completion criterion. Five repeats of a reference pose meet tolerance; unknown calibration versions are rejected.

Typical mistakes. Never reuse calibration after replacing a tendon without verification.

Session 12 Smooth Trajectory Generator

Hours 56–60 • Block 4

Session outcome

A motion editor and player with velocity and acceleration constraints.

Theory 60 minutes

- key poses and interpolation
- velocity profile
- derivative continuity
- phase-delayed wave

Practice 180 minutes

1. Define a versioned motion-file format.
2. Start each transition from measured state.
3. Create sway, circle, S-curve, wave, and return.
4. Plot all six channels and derivatives.
5. Run first without motors, then at low speed.

Measurement and report 60 minutes

Retain	Purpose
motion YAML files	evidence of completion and input to the next session
planner.py	evidence of completion and input to the next session
trajectory plots	evidence of completion and input to the next session
five-motion video	evidence of completion and input to the next session

Completion criterion. Commands contain no steps and the limiter rejects unsafe trajectories with a reason.

Typical mistakes. Never jump to a motion file’s first recorded point.

Session 13 ROS 2 Integration

Hours 61–65 • Block 5

Session outcome

A ROS 2 workspace for input, planning, limiting, bridging, and telemetry.

Theory 60 minutes

- topics, services, and actions
- QoS
- action cancellation
- host versus real-time control

Practice 180 minutes

1. Define command, state, and event messages.
2. Connect a gamepad through tail_input.
3. Send limited commands through tail_bridge.
4. Publish position, current, and state.
5. Test playback cancellation and planner loss.

Measurement and report 60 minutes

Retain	Purpose
ros2_ws	evidence of completion and input to the next session
interface definitions	evidence of completion and input to the next session
scenario rosbag	evidence of completion and input to the next session
node graph	evidence of completion and input to the next session
fault log	evidence of completion and input to the next session

Completion criterion. Embedded stopping works even when a ROS node exits; old actions never resume after STOPPED.

Typical mistakes. Do not move the fast protective loop into ROS.

Session 14 Endurance and Fault Testing

Hours 66–70 • Block 5

Session outcome

A 100-cycle protocol and controlled-failure report.

Theory 60 minutes

- test scenario and expected result
- fault observability
- thermal regime
- stop criterion

Practice 180 minutes

1. Run 100 slow wave cycles with logging.
2. Inspect tendons every 25 cycles.
3. Measure current and temperature consistently.
4. Simulate link loss, excess command, and a soft jam.
5. Repeat neutral and reference pose after endurance.

Measurement and report 60 minutes

Retain	Purpose
test_protocol.md	evidence of completion and input to the next session
100-cycle.csv	evidence of completion and input to the next session
wear photos	evidence of completion and input to the next session
fault matrix	evidence of completion and input to the next session
summary plot	evidence of completion and input to the next session

Completion criterion. No progressive damage appears and every injected fault reaches its defined response.

Typical mistakes. Stop on unusual noise, heat, or damage; record any configuration change.

Session 15 Wearable Test and Final Demonstration

Hours 71–75 • Block 5

Session outcome

Tail v1.0 with a complete evidence package and v2 backlog.

Theory 60 minutes

- staged energy increase
- human-in-the-loop stopping
- repeatable demonstration
- post-project review

Practice 180 minutes

1. Validate the mount on a dummy or equivalent load.
2. Wear the unpowered mechanism and assess mass and release.
3. Run one channel, one section, then the full slow sequence.
4. Demonstrate five reference motions and E stop.
5. Package the repository, report, BOM, logs, video, and launch instructions.

Measurement and report 60 minutes

Retain	Purpose
tail_v1.0 release	evidence of completion and input to the next session
final report	evidence of completion and input to the next session
demo video	evidence of completion and input to the next session
BOM	evidence of completion and input to the next session
known issues	evidence of completion and input to the next session
v2 backlog	evidence of completion and input to the next session

Completion criterion. The demonstration repeats from a clean environment and every criterion has saved evidence.

Typical mistakes. Never raise speed for a more dramatic video or claim safety outside validated modes.

Detailed Reading Guide

Appendix to the 75 hour build plan

This appendix maps specific books, papers, standards, and official documentation to every project session. It is deliberately selective: read the named chapters or topics, then apply them immediately to the tail. The complete recommended reading load is approximately 20–25 hours in addition to the 75 hours of engineering work; the minimum path is about 8–10 hours.

Core reference shelf

- **Lynch and Park, Modern Robotics** — Chapters 2–5, 8, 9, and 11: configuration, rigid transformations, kinematics, statics, dynamics, trajectories, and control. Open source
- **Webster and Jones, Design and Kinematic Modeling of Constant Curvature Continuum Robots** — Continuum architectures, constant-curvature coordinates, section frames, and model limitations. Open source
- **Rao et al., How to Model Tendon-Driven Continuum Robots and Benchmark Modelling Performance** — Tendon routing, modelling assumptions, gravity, friction, external loads, and model selection. Open source
- **Astrom and Murray, Feedback Systems** — Feedback principles, modelling, stability, PID, saturation, and discrete control. Open source
- **Budynas and Nisbett, Shigley's Mechanical Engineering Design** — Use as a reference for stress, bending, springs, fatigue, shafts, fasteners, and safety factors.
- **Scherz and Monk, Practical Electronics for Inventors** — Use as a reference for motor power, H-bridges, PWM, encoders, protection, filtering, and grounding.

How to use the list

1. Before a session, read only the required items and write five technical notes.
2. During the build, record which assumption each experiment tests.
3. After the session, save one calculation, plot, drawing, or test result derived from the reading.
4. Treat standards as design guidance, not as evidence that the prototype is certified.

Reading for Session 1

Measurable Requirements

Required reading

- **NASA Systems Engineering Handbook** — System requirements, functional and performance requirements, verification, validation, and traceability. Open source
- **Modern Robotics Chapter 2** — Configuration space, degrees of freedom, constraints, task space, and workspace. Open source

Concepts to understand

- goal versus requirement versus preference
- metric, tolerance, and acceptance threshold
- functional versus performance requirement
- verification versus validation
- traceability from requirement to saved evidence

Evidence after reading

Create requirements_v01.md with Requirement → Metric → Test → Acceptance threshold. Replace statements such as “moves naturally” with measurable limits for angle, rate, acceleration, pose-hold duration, and repeatability.

Reading rule

Do not postpone the build until every source is finished. Read the listed topic, write a concise engineering note, apply it in the session, and record where the physical result differs from the model.

Reading for Session 2

Risk and Safe Architecture

Required reading

- **ISO 12100** — Hazard identification, risk estimation, inherently safe design, guards, information for use, and residual risk. Open source
- **IEC 60204-1** — Electrical equipment of machines, supply isolation, protective bonding, and emergency stopping principles. Open source
- **ISO 13849-1 overview** — Safety-related control functions and the idea that risk reduction must be designed and validated. Open source

Concepts to understand

- pinch point and impact zone
- stored elastic energy
- single point of failure
- safe state and latched fault
- independent E-stop power removal
- unexpected-start prevention

Evidence after reading

Produce a hazard register with hazard, cause, severity, likelihood, prevention, detection, response, and residual risk. This study does not certify the prototype.

Reading rule

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Reading for Session 3

Kinematic Model in Python

Required reading

- **Modern Robotics Chapters 3–5** — Rotations, homogeneous transforms, forward kinematics, Jacobians, and statics. Open source
- **Webster and Jones** — Constant-curvature representation and transforms between sections. Open source
- **Open Continuum Robotics** — Worked constant-curvature model for a tendon-driven continuum robot. Open source

Concepts to understand

- local and world frames
- curvature kappa, bending angle theta, and bending-plane angle phi
- homogeneous transform composition
- zero-curvature limit
- assumptions omitted by the geometric model

Evidence after reading

Implement and test a three-section centreline model. Save neutral, axial, diagonal, circular, and phase-delayed wave plots with units and frame conventions.

Reading rule

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Reading for Session 4

Parametric Vertebra

Required reading

- **Autodesk Fusion learning collection** — Parametric sketches, user parameters, components, patterns, drawings, and manufacturing export. Open source
- **Shigley's Mechanical Engineering Design** — Stress concentration, bending, fatigue, fillets, fasteners, and factor of safety.
- **Creality CR-10 SE support material** — Printer setup, material profiles, calibration, and maintenance for the available machine. Open source

Concepts to understand

- fully constrained sketch
- parameter dependency
- FDM hole compensation
- layer-direction strength
- guide radius and edge finishing
- STEP versus STL

Evidence after reading

Publish three pitch variants, a dimension table, print settings, and a test chain. Measure the printed tendon passage and update the CAD compensation.

Reading rule

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Reading for Session 5

Passive Section

Required reading

- **Webster and Jones** — Continuum architectures, segmented backbones, compliant structures, and constant-curvature assumptions. Open source
- **Shigley's Mechanical Engineering Design** — Beam deflection, elastic energy, springs, creep, repeated loading, and fatigue.
- **Material data sheets** — Read the actual PETG, TPU 95A, fibreglass rod, and spring-strip data supplied by the chosen manufacturers.

Concepts to understand

- bending stiffness EI
- elastic return and neutral shape
- minimum bend radius
- hysteresis and creep
- stiffness versus safe compliance
- mass distribution

Evidence after reading

Compare at least three backbone configurations with the same geometry. Record mass, unloaded neutral, tip deflection, bend radius, and recovery after ten cycles.

Reading rule

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Reading for Session 6

Spool and Closed Tendon Loop

Required reading

- **Rao et al.** — Tendon routing, tendon-length mapping, friction, actuation space, and model assumptions. Open source
- **Capstan equation reference** — Understand tension multiplication caused by friction and total wrap angle. Open source
- **Selected tendon data sheet** — Rated load, construction, diameter, stretch, creep, bend radius, termination, and abrasion.

Concepts to understand

- antagonistic pair and closed loop
- pretension and slack
- spool radius and effective radius change
- Bowden friction
- hysteresis by approach direction
- tendon creep

Evidence after reading

Verify delta L = r delta alpha experimentally. Save pretension settings and a bidirectional command-versus-shape plot; reject routing that produces uncontrolled spool layering.

Reading rule

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Reading for Session 7

One Actuator Channel

Required reading

- **Selected motor, encoder, and driver data sheets** — Use continuous ratings, absolute maximum ratings, thermal limits, signal levels, and fault behaviour.
- **maxon Motor Data and Formulae** — Torque, speed, current, motor constants, gear ratio, efficiency, and thermal limits. Open source
- **ESP-IDF MCPWM** — Timers, comparators, generators, capture, synchronization, faults, and brake actions. Open source

Concepts to understand

- continuous versus stall torque
- torque = tendon force × spool radius
- gear ratio and efficiency
- encoder resolution at the spool
- current as load evidence
- hardware and software limits

Evidence after reading

Create a channel calculation sheet and low-energy test log. Confirm encoder sign, PWM sign, stop behaviour, idle current, loaded current, and voltage sag before attaching a tendon.

Reading rule

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Reading for Session 8

One Active Two Axis Section

Required reading

- **Feedback Systems** — Feedback loop, stability, proportional and integral action, disturbances, saturation, and anti-windup. Open source
- **Modern Robotics Chapter 11** — Motion control and feedback control of robot systems. Open source
- **Rao et al.** — Actuation-space mapping and tendon coupling. Open source

Concepts to understand

- closed-loop position control
- P, PI, and PID behaviour
- saturation and integral windup
- deadband and rate limiting
- mechanical cross-coupling
- controller sample time

Evidence after reading

Tune one loop at a time, then run a 3 × 3 two-axis command grid. Record both commands, both measurements, current, settling behaviour, and coupling instead of hiding mechanical defects with high gains.

Reading rule

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Reading for Session 9

Firmware and Fault Transitions

Required reading

- **The Rust Programming Language Chapters 6, 9, 11, and 16** — Enums, error handling, tests, and concurrency. Open source
- **The Embedded Rust Book** — Bare-metal development model, memory, exceptions, interrupts, and peripheral access. Open source
- **ESP-IDF Watchdogs** — Task watchdog, interrupt watchdog, configuration, and failure behaviour. Open source

Concepts to understand

- finite-state machine and legal transition
- latched fault and first-cause preservation
- monotonic time
- command freshness
- safe initialization
- fixed-capacity telemetry buffer

Evidence after reading

Build a transition table and unit tests for BOOT, READY, HOMING, MANUAL, PLAYBACK, STOPPED, and FAULT. Power-up and duplicate packets must never restart or refresh stale motion.

Reading rule

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Reading for Session 10

Three Section Mechanical Assembly

Required reading

- **Shigley's Mechanical Engineering Design** — Combined loading, bending moment, shafts, supports, fasteners, fatigue, and serviceability.
- **Rao et al.** — Multi-section routing and effects of friction, gravity, external loads, and coupling. Open source
- **NASA Systems Engineering Handbook** — Interface management and configuration management. Open source

Concepts to understand

- cascaded distal load
- base bending moment
- routing interfaces
- actuator placement near the body
- replaceable modules
- configuration-controlled assembly

Evidence after reading

Create a mass and moment budget for each section and the base. Freeze a routing drawing and photograph all boundary poses before applying motor power.

Reading rule

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Reading for Session 11

Six Channel Calibration

Required reading

- **NumPy linear algebra** — Matrix multiplication, least-squares solution, pseudoinverse, conditioning, and residuals. Open source
- **SciPy least_squares** — Bounded nonlinear least-squares fitting and residual analysis. Open source
- **NIST Technical Note 1297** — Measurement result, uncertainty components, repeatability, and reporting. Open source

Concepts to understand

- zero offset and homing
- calibration and coupling matrices
- repeatability versus reproducibility
- residual and outlier
- hysteresis by approach direction
- calibration versioning

Evidence after reading

Save raw CSV data, fit coefficients, residual plots, and a versioned YAML file bound to the CAD revision, tendon set, pretension method, spool radius, and firmware commit.

Reading rule

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Reading for Session 12

Smooth Trajectory Generator

Required reading

- **Modern Robotics Chapter 9** — Point-to-point trajectories, cubic time scaling, and quintic time scaling. Open source
- **SciPy CubicSpline** — Piecewise cubic interpolation, boundary conditions, and derivative evaluation. Open source
- **Ruckig documentation** — Online trajectory generation with position, velocity, acceleration, and jerk constraints. Open source

Concepts to understand

- position, velocity, and acceleration continuity
- jerk
- cubic versus quintic profile
- phase shift
- sample frequency and aliasing
- start from measured state

Evidence after reading

Plot all six channels and their first two derivatives. Create sway, circle, S-curve, travelling wave, curious-tip, and neutral-return motions; reject any file that violates validated limits.

Reading rule

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Reading for Session 13

ROS 2 Integration

Required reading

- **ROS 2 Jazzy documentation** — Nodes, topics, services, actions, parameters, launch, interfaces, and rosbag2. Open source
- **ROS 2 QoS concepts** — History, depth, reliability, durability, deadline, lifespan, and liveliness. Open source
- **RViz and URDF tutorials** — Robot model, robot_state_publisher, transforms, markers, and command-versus-measurement visualization. Open source

Concepts to understand

- topic, service, and action semantics
- action cancellation
- custom message versioning
- QoS compatibility
- JointState and diagnostics
- host planning versus embedded protection

Evidence after reading

Record a rosbag that contains desired pose, limited command, measured position, current, state, and fault events. Verify that the embedded controller stops safely when any ROS node exits.

Reading rule

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Reading for Session 14

Endurance and Fault Testing

Required reading

- **NASA Systems Engineering Handbook** — Verification, validation, test planning, configuration control, and anomaly reporting. Open source
- **ISO 12100** — Risk reduction verification and residual-risk review after design changes. Open source
- **ESP-IDF diagnostics** — Reset reason, watchdog behaviour, and fault evidence for embedded tests. Open source

Concepts to understand

- test scenario and expected result
- controlled fault injection
- stop criterion
- thermal regime
- progressive wear
- configuration change during testing

Evidence after reading

Run 100 slow wave cycles with inspections every 25 cycles. Save temperature, current, voltage, position, fault events, wear photographs, and before-versus-after reference-pose error.

Reading rule

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Reading for Session 15

Wearable Test and Final Demonstration

Required reading

- **ISO 12100** — Final hazard review, safeguards, warnings, and residual risks. Open source
- **ISO 13482 scope and principles** — Safety ideas for personal closely interacting personal-care robots; use as context, not a certification claim. Open source
- **NASA Systems Engineering Handbook** — Validation in the intended use scenario and evidence-based project closeout. Open source

Concepts to understand

- staged energy increase
- dummy-load validation
- breakaway mount
- observer-accessible stop
- repeatable demonstration
- known limitations and v2 backlog

Evidence after reading

Package CAD, BOM, firmware, Python, ROS 2 interfaces, calibration, logs, test report, known issues, and video. The tail must not be described as a balance aid, protective device, orthosis, or certified wearable robot.

Reading rule

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